World Robot Olympiad – Future Engineers

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1. Introduction & Objectives

1.1 Purpose of the Documentation

The purpose of this document is to provide a clear and detailed guide to developing and constructing an autonomous robotic vehicle for the World Robot Olympiad (WRO) – Future Engineers competition.

It contains all relevant information about the components used and their specifications, enabling readers to understand and reproduce the project.

It also explains the challenges encountered during development and the solutions implemented to overcome them.

All design and component choices are supported by reasons and technical arguments. Finally, the results achieved are presented.

1.2 Project Overview

The aim of the project is to develop a fully autonomous vehicle capable of navigating a course independently, performing manoeuvres based on obstacles, and completing the task quickly and without errors.

Achieving this requires robust control logic that integrates multiple sensors with electrical and mechanical systems.

1.3 About the Team and Its Formation

There is a very particular energy that fills a laboratory at 10 p.m. when you are trying to solve a logic bug that should not exist in the first place. That energy defined our entire semester. We did not choose this competition because we had plenty of free time; rather, during the first weeks of our robotics studies at the Technical University of Applied Sciences Würzburg-Schweinfurt (THWS), we realised that theory alone can only take us so far. We wanted to experience what happens when our own code actually makes something move in the real world.

This project was created to give students on the THWS Bachelor's degree in Robotics the opportunity to apply theoretical knowledge in a practical, competition-oriented engineering environment. Our team consists of three second-semester Robotics students who share a strong interest in autonomous systems and intelligent machines. The idea of entering the World Robot Olympiad (WRO) in the Future Engineers category was proposed by Professor Fabian Dax, who had previously served as a judge at WRO competitions. Based on his experience, he recognised that the competition offered an excellent opportunity to gain practical experience with autonomous driving systems and robotic construction.

Our journey began later than that of many other teams. While others were already building their first prototypes, we were still developing our basic concept. In retrospect, this became one of our greatest advantages because it forced us to make every technical decision deliberately. Motivated by our shared interest in robotics and programming, we formed a team and accepted the challenge. Under the guidance of Mr Fabian Dax, who supported us as a mentor throughout the project, and Professor Reiner Herrler, we gained access to the robotics laboratories of the Center for Robotics (CERI), the robotics department at THWS in Schweinfurt.

Throughout the project, the university supported us by providing access to 3D printers, soldering stations, test environments, workshop tools, and financial assistance for building the autonomous vehicle. We spent many days—and quite a few nights—in these laboratories, surrounded by prototypes, sensors, cables, and constant experimentation. We did not just build a robot; we also learned what being engineers really means: analysing problems, testing solutions, accepting mistakes, and starting again.

The project was also closely connected to several courses from our first semester, especially those covering microcontrollers, programming, and electronics. It allowed us to improve our practical skills and deepen our understanding of sensors and their integration into a real autonomous system.

What makes our team special is that we do not all think alike. Raghav Narendra Patil approaches robotics primarily from the perspectives of mechanics, RC systems, and industrial performance. Vineet Mukund Khare is particularly interested in bio-inspired robotics and autonomous systems such as drones, where efficiency and movement dynamics are central. Icha focuses mainly on machine learning and humanoid robotics—specifically, how machines can become more intelligent and adaptable. These different perspectives did not always lead to quick decisions, but they ensured that every solution was considered and improved from several angles.

This project included many “back to the drawing board” moments, but that was precisely where its greatest value lay. It taught us that passion does not mean starting perfectly prepared; it means having the determination to continue despite time pressure, setbacks, and uncertainty until a solution finally becomes tangible.

2. Development Process

2.1 Project Timeline

The project began with planning and brainstorming. Together with our mentor, the team created a schedule and made initial decisions about the drive configuration, sensor selection, and obstacle detection. Documentation also began at this stage. We then compared different chassis, sensors, and microcontrollers. Following a cost and feasibility analysis, we selected the WRO Future Engineers Set 2026 because it suited our needs.

Next came construction. We first assembled the chassis, then optimised the drive and steering and integrated mechanical components such as axles, gears, bearings, and the steering mechanism. During the electronics phase, we designed circuits, checked components, and tested and validated the sensors using breadboards and soldered connections.

Sensor positioning was the longest phase. We tried different arrangements and iteratively adjusted the mechanical design, programming, and solutions to voltage problems; some sensors had to be replaced. In parallel, the software was developed through several iterations, supported by flowcharts, logic diagrams, and extensive debugging to ensure stable operation.

During testing, we built a simple test course and repeatedly tested the vehicle. This revealed necessary changes to the sensors, software, and system architecture, and these changes were implemented; additional sensors were integrated when needed. This iterative process ultimately led to stable overall performance, with continuous improvements throughout.

2.2 Milestones

The project began on 25 February 2026, with participation planned for 26 April 2026.

First functional version (5 March 2026): The vehicle could detect obstacles and avoid them by moving both forwards and backwards.

First successful navigation (20 March 2026): The vehicle completed the course with approximately 50% reliability.

First major setback (23 March 2026): All sensors were damaged because of voltage problems, forcing the project to pause until 2 April 2026.

First colour detection (7 April 2026): The vehicle could detect coloured fields, but could not yet perform effective manoeuvres.

First redesign (completed by 22 April 2026): The vehicle was redesigned and optimised through several iterations.

First successful manoeuvre around coloured fields (24 April 2026): Successful navigation and manoeuvres around coloured fields were achieved.

First draft of the documentation (24 April 2026): The initial documentation was completed.

Videos uploaded (25 April 2026): The recorded videos were uploaded on 25 April 2026.

Complete documentation draft (25 April 2026): The first complete draft of the documentation was finished.

Improvements made: Continuous optimisation was carried out to further improve performance.

3. Technical Specifications

This section lists only the names of the main components. For the technical specifications of the individual components, please see Appendix A.

3.1 General Specifications
- **Robot weight: 0.955 kg**
- **Dimensions (L × W × H): 28 × 15.5 × 13.5 cm**
- **Length: 28 cm**
- **Width: 15.5 cm**
- **Height: 13.5 cm**

3.2 Battery
- **Device/type: 1800 mAh rechargeable LiPo 25C battery (Conrad Energy), quantity: 1**

3.3 Microcontrollers
- **ESP32 Dev Kit V1 (AZ-Delivery), quantity: 1**
- **Arduino Nano V3.0, quantity: 1**

3.4 Motor Specifications
- **Steering motor: MG996R servo**
- **Drive motor: GA25-370 DC geared motor (6 V, 60 RPM)**

3.5 Sensor Specifications
- **Camera: Pixy2 (version 2.1), quantity: 1**
- **HC-SR04 ultrasonic distance sensor, quantity: 3**
- **Inertial Measurement Unit (IMU): MPU6050, quantity: 1**

4. Mobility and Mechanics

This section covers the drive system, steering system, and their associated mechanisms.

Figure 2: Rear view of the vehicle showing (1) the GA25-370 DC geared motor, (2) the gear transmission system (with gear ratios of 1:18.9 or 1:61.2), (3) the rear wheels, and (4) the rear axle.

Figure 3: Top view of the steering system showing (1) the MG996R servo motor, (2) connecting rod 1 (connecting the servo to the right wheel), (3) a plastic steering knuckle, (4) connecting rod 2 (connecting the right wheel to the left wheel), and (5) the left plastic steering knuckle.

The robot uses a two-motor configuration: a DC geared motor for forward and reverse movement, and a servo motor for steering. The drive motor is a GA25-370 DC geared motor (6 V, 60 RPM), shown in Figure 2 (1). It is mounted at the rear of the vehicle and is responsible for forward and reverse movement. It drives the rear axle through a gear transmission system, shown in Figure 2 (2), which connects via the rear axle (Figure 2 (4)) to both rear wheels (Figure 2 (3)). Power is transmitted through a gear system with two available ratios, 1:18.9 and 1:61.2. A smaller gear is mounted on the motor shaft, while a larger gear is connected to the rear axle, enabling efficient torque transfer to both wheels. The rear drive motor is therefore connected directly to the wheels through this gear system, providing reliable power transmission during operation.

For steering, an MG996R servo motor (Figure 3 (1)) is installed near the front centre of the chassis. The servo is connected to a system of levers and joints that allows both front wheels to steer simultaneously. One lever (Figure 3 (2)) is connected directly to the wheel joint (Figure 3 (3)), while a second lever (Figure 3 (4)) connects to the opposite steering linkage (Figure 3 (5)). Together, they form an Ackermann steering system (Figure 4), ensuring that both wheels turn together. The levers consist of metal rods with ball joints at their ends to provide smooth movement with low friction. This steering mechanism improves cornering because the inner and outer wheels can turn through slightly different angles. The drivetrain is therefore classified as rear-wheel drive (RWD) with Ackermann steering.

This configuration was chosen because it is one of the simplest possible vehicle designs and provides both mechanical and electrical simplicity. At the same time, the system is reliable and fully compatible with the competition rules, which limit the number and type of motors that may be used. It also uses inexpensive, widely available components, making repairs, maintenance, and replacement of spare parts straightforward and economical.

5. Power and Sensors

5.1 Power

Figure 4: Side view of the electronic components: (1) Conrad Energy LiPo battery, (2) ON/OFF rocker switch, (3) HW-688 buck converter, (4) L298N H-bridge motor driver, (5) XT60 connectors, (6) resistors for current limiting and signal conditioning, (7) various jumper wires for electrical connections, and (8) the breadboard as the central wiring integration platform.

The technical specifications of all components used can be found in Appendix B.

The robot is powered by a rechargeable 1800 mAh 25C LiPo battery from Conrad Energy (Figure 4 (1)), which supplies the entire system. Its high discharge capability, low weight, and stable voltage output make it suitable for mobile robotics applications. A rectangular ON/OFF rocker switch (Figure 4 (2)) is installed between the battery and the circuit, allowing the whole system to be switched on and off safely. Since different components require different operating voltages, an HW-688 buck converter (Figure 4 (3)) is used for voltage regulation. It reduces the battery voltage from 11.1 V to a stable 5 V supply for the microcontrollers and sensors, ensuring reliable operation.

Several resistors (Figure 4 (6)) are used for electrical signal conditioning. Three 220 Ω resistors limit current through the LEDs, and two 4.7 kΩ resistors are used for I²C communication between the Arduino and ESP32. In addition, voltage dividers made from 1 kΩ and 2 kΩ resistors reduce the ultrasonic sensors' 5 V output signal to 3.33 V, ensuring compatibility with the ESP32. A 220 µF capacitor is connected in parallel with the power supply to smooth voltage spikes caused by sudden current draw from the servo motor and improve system stability.

Two modified solderless breadboards are used for power distribution and act as central distribution points. The buck converter's 5 V output is distributed through them to the Arduino Nano, ESP32, and all sensors. Both the 5 V supply and ground are shared throughout the system. The motor system is powered separately through an L298N H-bridge motor driver (Figure 4 (4)), connected in parallel with the main power supply. This gives the driver the full 11.1 V battery voltage and allows it to control the DC motors' speed and direction. XT60 connectors (Figure 4 (5)) and various jumper wires (Figure 4 (7)) provide electrical connections. The breadboards (Figure 4 (8)) serve as the central integration platform and allow orderly wiring.

Figure 5 shows the power distribution, or energy flow, in the robot. The 11.1 V LiPo battery is the main energy source. After the ON/OFF switch, the power supply splits into two branches: the first powers the microcontrollers and sensors at a stable 5 V through the buck converter, while the second supplies the L298N motor driver directly for driving the motors. This provides a stable and clearly organised power supply for the complete system.

5.2 Sensors

Figure 5: Sensor arrangement showing (1) the Pixy2 camera (version 2.1), (2) the HC-SR04 ultrasonic sensors, and (3) the MPU6050 Inertial Measurement Unit (IMU).

The robot uses a combination of camera, distance, and orientation sensors for reliable autonomous navigation. A Pixy2 camera (version 2.1) (Figure 5 (1)) is used for colour-based obstacle detection. It is connected directly to the Arduino through the ICSP pins, where the detected data are processed and the required steering angle is calculated before being sent to the ESP32.

Three HC-SR04 ultrasonic sensors (Figure 5 (2)) are used for wall detection, obstacle avoidance, and detection of corner areas. They are mounted at the front-left, front-right, and centre-front positions of the vehicle to cover the front and sides. The sensors are connected to the ESP32 and powered by the 5 V supply. Since the ESP32 uses a 3.3 V logic level, voltage dividers made from 1 kΩ and 2 kΩ resistors reduce the sensors' 5 V output signal to a safe 3.33 V. An MPU6050 Inertial Measurement Unit (IMU) (Figure 5 (3)) measures yaw, meaning rotation around the vertical axis. This data is used for steering corrections and stabilisation while driving.

The sensor positions were carefully selected to improve detection accuracy and response time. The Pixy2 camera is mounted on a 3D-printed bracket 10 cm above the chassis and tilted downwards by 60°. This elevated position provides a clear view of the course, avoids detecting irrelevant background objects, and ensures that only relevant course elements are visible. It also leaves enough room for the front ultrasonic sensors. The three ultrasonic sensors are mounted at a height of 4 cm on the front bumper using adjustable 3D-printed brackets. One sensor points forwards, while the other two point left and right. This arrangement provides almost complete coverage of the front and

side areas for obstacle detection. The side sensors were deliberately placed on the front bumper rather than in the middle of the vehicle, because a central position caused delayed obstacle detection and slower turning responses. Placing them at the front allows detection as soon as the bumper reaches an obstacle, significantly improving steering accuracy. The MPU6050 is mounted flat on the rear breadboard and aligned with the direction of travel. Since only yaw movement needs to be measured, its exact position is less important than correct alignment. This placement enables accurate angular-velocity measurements and stable heading control during navigation.

This sensor arrangement was chosen to minimise delays in obstacle detection and improve steering response. The camera improves colour detection, the ultrasonic sensors enable rapid obstacle detection, and the IMU provides stable heading corrections. Although no rear sensors are used—creating some blind spots, particularly at the rear and in the central diagonal areas—the system remains reliable and fully adequate for the competition requirements.

Figure 6: Block diagram of the robot’s electronic circuit.

6. Obstacles & Navigation

The code for both of the robot’s microcontrollers is available in the GitHub repository:
https://github.com/V-yeet/WRO---THWS-VDE-SCHWEINFURT-TEAM.git

Video: https://youtu.be/mlXUz-9Uu8A

6.1 Cornering / Path Following

The robot navigates corners by following walls and automatically turning at corners. Two ultrasonic sensors measure distances to the walls, while a PD controller (Kp = 4, Kd = 4) keeps the robot centred between them. The nearer wall is used as the reference. A corner is detected when the front distance is less than or equal to 60 cm, one side reports a very large distance above 255 cm, and the other side reports less than 100 cm. In this case, the robot steers towards the open side. The turn is monitored using the MPU6050 and ends once the yaw angle has changed by approximately 80°. The heading is then set to 90°.

6.2 Orientation & Boundary Detection

The robot’s orientation is based entirely on the MPU6050 yaw angle, which is transmitted over a serial connection. This value is used both to monitor turns and to update the target heading. Three ultrasonic sensors are used to detect boundaries. If an obstacle is less than 7.5 cm in front of the front sensor, the robot reverses to avoid getting stuck. The side sensors also support wall following. Although the camera can detect walls, it is not used for boundary detection.

6.3 Obstacle Detection

The PixyCam detects obstacles using colour markers. Red and green objects are stored as separate signatures. An obstacle is considered relevant as soon as its y-coordinate in the camera image exceeds a threshold of approximately 40, indicating that it is sufficiently close.

6.4 Obstacle Avoidance

When an object is detected, the robot interrupts wall following and first aligns itself with the object using its x-coordinate until the object is centred in the camera image. As the object approaches and reaches a y-coordinate of approximately 70, a lateral offset is introduced into the control system. This causes the robot to move a few centimetres sideways and go around the obstacle. Depending on the object’s colour, the robot moves left or right and then reorients itself using the nearest wall as a reference. The offset remains in effect even after the object disappears from view, ensuring stable continued movement.

6.5 Programming Challenges

Initially, we tried counting corners to determine the direction of travel and the number of laps completed. A colour sensor was used for this, but it required reliable calibration using black and white references. Since the course only had white available, this method proved unreliable and was abandoned.

We then used the camera to detect lines. The lines could be detected by mounting the camera higher and at an angle, but later integration of obstacle detection meant that the lines either remained visible for too long or were no longer detected correctly. This approach was also abandoned. Finally, corner detection was implemented using ultrasonic

sensors. A corner is detected when a sensor measures a very large distance, indicating an open direction. The robot then turns until the yaw angle changes by approximately 80°. This method proved much more reliable. Determining the direction of travel by counting corners was also error-prone because the counter increased multiple times within the turning loop. Instead, the yaw angle was used directly and rounded to the nearest multiple of 90°. This made heading determination more stable.

Lap counting was also initially based on corners, but this proved unreliable. The total change in yaw is now used instead. Three laps correspond to approximately 1080°. The robot stops once this value is reached and the front sensor simultaneously indicates that it is close to the starting point again. During obstacle avoidance, the robot tended to oversteer. The combination of the PD controller and heading stabilisation is not yet optimally tuned and requires further adjustment.

6.6 Code & Explanation

In the ESP32 program, all variables are defined first. At the start of the main loop, the program checks whether a connection to the Arduino Nano exists. Only then does the program run fully. The distance to the wall is also stored so that the robot can stop in the correct position after three laps. The maximum rotation is limited to approximately 1040°. To prevent collisions, the robot reverses if sensor readings fall below a threshold. Detected obstacles are also stored as Boolean variables. At the start, the direction of travel is determined using the side sensor readings.

When a coloured block is detected, the robot uses camera data to align itself with the object. Once the object is close enough, it switches to avoidance mode, in which the wall-distance targets are adjusted. Steering is controlled by a PD controller supplemented by the target heading to ensure stable straight-line driving. The yaw angle is rounded to multiples of 90°. Different control modes combine these control methods according to the situation. A turn is triggered when the sensors detect an open direction. During the turn, the yaw angle is monitored continuously. Once the maximum total rotation is reached, the robot stops at the previously determined position.

In the Arduino Nano program, all variables required for sensors and calculations are defined. The MPU6050 is calibrated at startup to reduce measurement errors. Acceleration and angular velocity are then measured continuously and integrated over time to calculate velocity and yaw angle. Small measurement deviations are filtered out. At the same time, the PixyCam detects up to three objects per cycle and transmits their positions and colours. All data are sent to the ESP32 in a fixed format every 50 milliseconds to enable real-time control.

7. Complete Robot System and Technical Decisions

Figures 7–14: Front, top, underside, right-side, rear, and left-side views of the robot. Figure 14 shows the mechanical construction and system components, numbered as follows: (1) YFROBOT chassis kit, (2) aluminium base plate, (3) acrylic cover, (4) 3D-printed spacers, (5) ultrasonic sensors, (6) 3D-printed sensor mounts, and (7) angled camera mount.

7.1 Chassis

The YFROBOT Chassis Kit (Figure 14 (1)) was selected because it has Ackermann steering and rear-wheel drive, functioning like a conventional rear-wheel-drive vehicle. This makes the system much easier to control and integrate. The competition rules restrict the permitted motor configuration, and this chassis fully satisfies the requirement to use one motor for forward and reverse movement and a separate servo motor for steering. Another important reason for selecting this chassis is its structural stability and build quality. Since the team planned to expand the vehicle with additional 3D-printed brackets, sensor mounts, and support structures, the metal base provides a robust and stable foundation for these modifications. Furthermore, this chassis is included in the World Robot Olympiad (WRO) – Future Engineers Set 2026 (Funduino), making it a suitable main platform for the robot. The system uses front-wheel Ackermann steering with rear-wheel drive.

7.2 Mechanical Design

The mechanical design combines aluminium, acrylic, and 3D-printed parts into a stable and functional structure. The main materials and parts include an aluminium base plate, an acrylic cover for the motor, a raised platform, 3D-printed spacers and sensor mounts, an upper platform, a foam bumper, aluminium angle brackets, brass, iron, and stainless-steel washers, nuts and bolts, stainless-steel axles, rubber and plastic wheels, plastic gears, and stainless-steel steering linkages.

The chassis is assembled from the bottom upwards, beginning with the aluminium base plate (Figure 14 (2)), which has several mounting points for motors and structural brackets. Aluminium angle brackets securely attach both the servo motor and the DC drive motor to the base plate. Brass spacers raise an acrylic plate above the rear motor area. This acrylic plate (Figure 14 (3)) covers the drive motor and also serves as a mounting platform for the breadboards and electronic components. A second platform sits above it, supported by 3D-printed spacers (Figure 14 (4)) and positioned 10 cm above the base plate. This upper plate carries the angled camera mount for the Pixy2 sensor. A foam bumper is attached to the front of the base plate to protect the chassis and sensors during collisions. Three ultrasonic sensors are mounted on this bumper (Figure 14 (5)) using 3D-printed brackets (Figure 14 (6)). The servo motor is connected to a dual-link steering system attached to the wheel suspensions, forming the Ackermann steering mechanism. The DC drive motor is mounted on brackets and transfers motion through a gear system to the rear axle, with selectable gear ratios of 1:18.9 and 1:61.2. All electronic components are mounted on breadboards, allowing centralised wiring, easier maintenance, and simpler debugging.

7.3 Additional Components

Several additional components were added to improve functionality and protection. 3D-printed spacers support the raised platform, which sits 10 cm above the base plate. This platform carries a 3D-printed angled camera mount (Figure 8 (7)), which points the Pixy2 camera at the course at an angle of 60° to improve object detection. The camera position can be adjusted according to the desired field of view. It can be mounted either on the 60° angled bracket on the raised platform or on the 3D-printed bracket for the front ultrasonic sensors for a more direct forward view. This flexibility allows object and track detection to be optimised according to test conditions and competition requirements. Additional 3D-printed mounts hold the three ultrasonic sensors approximately 4 cm above the front bumper, pointing in the required directions (forward, left, and right). A foam bumper is also installed at the front to protect the chassis, wheels, and sensors from impacts and improve the robot’s durability during testing and competition.

7.4 Technical Decisions

This section describes some of the main problems the team encountered and the technical decisions made to address them. Decisions related to programming and logic are described in the “Code” section.

Insufficient voltage regulation — Incorrect voltage regulation and incorrectly selected resistor values damaged or destroyed all the ultrasonic sensors. We checked the sensor datasheets, calculated the required resistor values, and verified voltage levels using multimeters.

Lighting problems with the camera / colour sensor — The Pixy camera worked best under ideal lighting. When ambient lighting changed, it detected false objects (false positives) or sometimes failed to detect objects. LEDs were therefore installed near the camera and colour sensor to provide sufficient illumination for reliable object detection. However, this did not fully solve the problem. The LEDs were removed and the camera was tilted at an angle of 60° to the track, after which false positives were no longer detected. New corner logic that did not use a colour sensor was also developed.

Corner markings — Because of its mounting angle, the camera could not reliably detect the corner markings. A separate sensor was therefore used to detect the orange and blue corner lines. A colour sensor was mounted underneath the vehicle to detect these lines.

Communication between microcontroller boards (ESP32 and Arduino) — Initially, we attempted to implement communication over I²C, but it did not work as expected and was difficult to implement. We therefore switched to serial/UART communication. I²C required custom libraries and additional code developed through trial and error, making the system unnecessarily complicated. UART communication between the boards was simpler, easier to debug, and therefore the preferred solution.

Removing the colour sensor — The colour sensor had to be recalibrated every time the vehicle was switched on. Calibration required both a black and a white surface, which was impractical during the competition. The colour sensor was therefore removed, and the coloured lines in the corner sections were no longer used. Instead, ultrasonic sensors were used for corner detection.

Calibration — All sensors and the steering system had to be calibrated. None of these systems worked perfectly with the originally assumed ideal values. Through trial and error, constants were determined that enabled optimal behaviour from the sensors and steering system.

Changing the corner logic — Initially, the team attempted to use orange and blue lines to detect corner sections and turns. This approach was not suitable for our situation. The camera could not reliably detect the corner lines because

of its orientation, and the colour sensor had to be recalibrated at every startup, which was impractical. The method was therefore abandoned.

The final solution was to use the already installed ultrasonic sensors not only for obstacle detection and avoidance but also for wall following. The vehicle maintained a constant distance from the wall and detected corners when the orientation of the boundary wall changed.

Ultrasonic sensor placement — Initially, four ultrasonic sensors were arranged so that the first two were mounted at the front-left and front-right of the vehicle, angled at 45° to the centreline. This allowed obstacles outside the camera's field of view to be detected and covered the entire front area. The other two ultrasonic sensors were placed near the middle of the vehicle and pointed to the right and left.

This arrangement worked while the colour sensor was used to detect corner sections. However, once corner detection was transferred entirely to the ultrasonic sensors, this configuration proved ineffective and unnecessarily complicated. A new arrangement was implemented: one sensor pointed forwards and two pointed right and left. All three sensors were mounted on the front bumper.

Delayed turning caused by sensor placement — Two ultrasonic sensors were initially mounted near the middle of the vehicle. This meant obstacles and walls were detected too late, causing delayed turns and poor driving performance. The sensors were therefore moved to the front of the vehicle and oriented to the right and left so that the steering system could respond immediately to changes in distance.

7.5 Overview of the Complete Robot System

The complete robot system was developed with careful consideration of the relationship between chassis design, component selection, energy consumption, computing power, and overall weight. Choosing the YFROBOT chassis with Ackermann steering and rear-wheel drive provided a stable, realistic vehicle structure while satisfying the competition rules. Lightweight yet robust materials such as aluminium, plastic, and PLA were preferred to reduce energy consumption while remaining within the weight limit. Components including the ESP32, Pixy2 camera, ultrasonic sensors, MPU6050, and LiPo battery were selected for sensor accuracy and system reliability. A stable power supply, with suitable voltage regulation through a buck converter, resistors, and decoupling capacitors, improved performance and reduced electrical interference. Several design versions were tested and improved during development, including changes to sensor placement, steering calibration, voltage protection and regulation, and software logic. These changes improved obstacle detection, steering accuracy, and overall reliability, resulting in a more robust and competitive autonomous robot.

Video: https://youtu.be/mlXUz-9Uu8A

8. References

• Official Pixy2 CMUcam5 documentation: https://docs.pixycam.com/wiki/doku.php?id=wiki:v3:overview
• Pixy2 SPI communication guide: https://docs.pixycam.com/wiki/doku.php?id=wiki:v3:porting_guide
• Espressif Systems Arduino-ESP32 documentation: https://docs.espressif.com/projects/arduino-esp32/en/latest/
• Official Arduino Nano documentation: https://docs.arduino.cc/hardware/nano
• HC-SR04 distance sensor guide: https://randomnerdtutorials.com/complete-guide-for-ultrasonic-sensor-hc-sr04/
• ESP32 + MPU6050 complete guide: https://www.esp32s.com/blog/the-complete-esp32-mpu-6050-guide-master-motion-sensing-with-arduino-ide/
• World Robot Olympiad – Future Engineers 2026 rules: https://wro-association.org/competition/2026-season/

Appendices – Component Specifications

APPENDIX A – MOBILITY

Battery
- **Device/type: 1800 mAh rechargeable LiPo 25C (Conrad Energy)**
- **Quantity: 1**
- **Manufacturer: Conrad Energy**
- **Nominal voltage: 11.1 V**
- **Maximum discharge current: 90 A**
- **Maximum charging current: 3.6 A**
- **Capacity: 1800 mAh**
- **Nominal energy: 19.98 Wh**

Switch
- **Type: rectangular ON/OFF rocker switch**

Voltage regulation
- **Device: HW-688 buck converter**
- **Input: 9–36 V**
- **Output: 5 V**
- **Maximum current: 5 A**

Resistors
- **220 Ω (3×), 4.7 kΩ (2×), 1 kΩ (4×), 2 kΩ (4×)**

Voltage stabilisation
- **Capacitor: 220 µF**

Power distribution
- **Device: breadboards (×2), Shenzhen Hailege Technology Co., Ltd.**

Motor power supply
- **Device: L298N H-bridge motor driver**

Wiring
- **XT60 connectors**
- **Jumper wires (male–male, male–female, female–female)**
- **Breadboard dimensions: 20 × 15 mm**
- **Weight: 2–3 g**

APPENDIX B – SENSORS

Camera – Pixy2 (version 2.1)
- **Processor: NXP LPC4330, 204 MHz, dual-core**
- **Frame rate: up to 60 FPS**
- **Sensor model: Aptina MT9M114**
- **Resolution: 1296 × 976 pixels**
- **Communication: SPI**

Ultrasonic distance sensor – HC-SR04
- **Type: ultrasonic sensor**
- **Principle: time-of-flight measurement**
- **Measuring range: 2–400 cm**
- **Accuracy: ±3 mm**
- **Resolution: 0.3 cm**
- **Measuring angle: 15°**
- **Operating voltage: 5 V DC**
- **Current consumption: 15 mA**
- **Interface: GPIO (TRIG/ECHO)**
- **Pins: VCC, GND, TRIG, ECHO**
- **Dimensions: approximately 45 × 20 × 15 mm**
- **Weight: approximately 10 g**
- **Distance formula: Distance = (Time × 0.034) / 2**

Inertial Measurement Unit – MPU6050
- **Type: 6-axis motion sensor**
- **Use in this project: gyroscope only**
- **Manufacturer: InvenSense (TDK)**
- **Gyroscope range: ±250°/s**
- **Output rate: approximately 1 kHz**
- **Operating voltage: 5 V**
- **Current consumption: 3.6 mA**
- **Communication: I²C**

APPENDIX C – STRUCTURE & CHASSIS

Chassis
- **YFROBOT Chassis Kit**
- **Chassis dimensions: 250 × 148 × 64 mm**
- **Weight: 700 g**
- **Maximum load capacity: 1500 g**
- **Ground clearance: 14 mm**
- **Turning radius: 450 mm**
- **Gear ratios: 1:18.9 / 1:61.2**
- **Drive system: Ackermann steering (front steering, rear-wheel drive)**

Materials
- **Aluminium base plate**
- **Acrylic plates**
- **3D-printed parts**
- **Aluminium angle brackets**
- **Stainless-steel axles**
- **Nuts and bolts**
- **Washers (brass / stainless steel / iron)**
- **Rubber and plastic wheels**
- **Plastic gears**
- **Foam bumper**

Steering
- **MG996R servo**
- **Ackermann steering linkage**

Drive
- **GA25-370 DC geared motor (6 V, 60 RPM)**
- **Gear transmission**